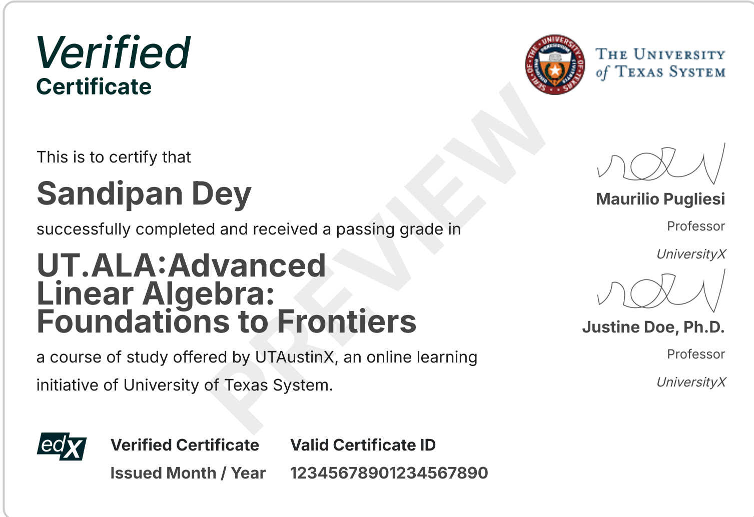
4.4.1 A has linearly independent columns

Bookmark this page

Pursue a verified certificate



- Unlimited access to all course materials
- Shareable verified certificate
- AI learning assistant and translations

Upgrade for $99

Week 04 due Feb 4, 2025 05:33 IST

Video

So we could instead

project onto the column space of Q. Okay? And that's the simpler formula. That's Q

Hermitian times b, sorry, ithat's Q times Q Hermitian times b. Why? Because the Q

Hermitian Q that ends up in the middle here that you the end of inverting, is just the

identity matrix. Now we can then also replace A by Q times R and what we find

is that Q times R times x-hat is equal to Q times Q Hermitian times b. Okay?

4:37 / 4:37

2.0x

Video

Download video file

Transcripts

Download SubRip (.srt) file

Download Text (.txt) file

Reading assignment

1/1 point (graded)

Read Unit 4.4.1 of the notes. [LINK]

- [x] done

Submit

Correct (1/1 point)

Discussion

Hide Discussion

Topic: Week 04 / 04.4.1

Add a Post

Show all posts | by recent activity

There are no posts in this topic yet.

Homework 4.4.1.1.

1/1 point (graded)

Yet another alternative proof for Theorem 4.4.1.1 starts with the observation that the solution is given by $\hat{x} = (A^H A)^{-1} A^H b$ and then substitutes in $A = QR$. Give a proof that builds on this insight.

- [x] done

Submit